Tidal Rhythms in Urban Beekeeping: Hive Thermoregulation Under Rooftop Heat Islands

Authors: Dr. Maren Okafor-Lindqvist, Dr. Tomas Verhoeven
Affiliation: Institute for Urban Pollinator Ecology, Lakeside University
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ABSTRACT

We studied 48 honeybee hives (Apis mellifera) across 12 rooftops in a dense coastal city over 26 months. Rooftop temperatures exceeded surrounding green areas by 4.1 degrees Celsius on average during summer. Colonies responded by increasing fanning labor, which reduced foraging time by 17 percent. Hives painted white and fitted with a 3 cm reed insulation layer kept brood temperature within 1.2 degrees of the optimal 34.5 degrees Celsius, while untreated hives drifted by up to 3.8 degrees. Honey yield in treated hives was 22 percent higher. We propose the Reed-and-Reflect protocol as a low-cost intervention for urban apiaries.

1. INTRODUCTION

Urban beekeeping has grown quickly in the last decade, yet most guidance is adapted from rural practice. Cities create heat islands caused by asphalt, concrete and reduced vegetation. Honeybees keep brood nests at a narrow temperature band of 34 to 35 degrees Celsius. Deviations cause developmental defects and higher larval mortality. Little is known about how rooftop placement, a common choice in cities, affects this balance.

2. METHODS

2.1 Site selection

We selected 12 rooftops between 8 and 40 meters above street level. Each rooftop hosted four hives. Two hives per rooftop were assigned to the Reed-and-Reflect treatment and two served as controls.

2.2 Sensors

Each hive carried five temperature sensors placed at the brood center, brood edge, honey super, entrance and exterior wall. Readings were logged every 10 minutes. Colony weight was measured on a hive scale once per hour. Foraging activity was counted with an optical entrance counter.

2.3 Treatment

The Reed-and-Reflect protocol has two parts. First, the hive exterior is painted with a white, non-toxic, water-based reflective coating. Second, a 3 cm layer of dried common reed (Phragmites australis) is fixed to the roof and side walls of the hive, leaving the entrance and ventilation slits open.

3. RESULTS

3.1 Temperature stability

In summer, control hives showed brood temperature deviations of up to 3.8 degrees Celsius from the optimum, with 14 percent of readings outside the safe band. Treated hives deviated by at most 1.2 degrees, with only 2 percent of readings outside the band.

3.2 Foraging and fanning

Control colonies devoted on average 31 percent of workers to fanning during afternoon peaks, compared with 18 percent in treated colonies. As a result, foraging time fell by 17 percent in controls relative to treated hives.

3.3 Honey yield

Over the second season, treated hives produced a mean of 31.4 kg of honey per hive, compared with 25.7 kg in controls, an increase of 22 percent. Winter surviva

l was 94 percent in treated hives and 81 percent in controls.

3.4 Effect of height

Rooftop height had a weaker effect than expected. Hives above 25 meters experienced stronger wind, which lowered peak temperatures by about 0.6 degrees Celsius, but also increased flight difficulty for foragers. The net effect on yield was not statistically significant ($p = 0.21$).

4. DISCUSSION

The results show that passive thermal management is an effective and cheap way to support urban colonies. The treatment cost approximately 12 euros per hive, mostly for reed and paint. Compared with active cooling systems, which need electricity and maintenance, the Reed-and-Reflect protocol is simple to adopt by hobbyist beekeepers.

Limitations include the single climate zone and the short reed lifetime of about 3 years before replacement. We did not test other insulation materials such as cork or straw, which may perform differently.

5. CONCLUSION

Rooftop heat islands impose a measurable burden on honeybee colonies. A white coating combined with a reed insulation layer reduces thermal stress, preserves foraging time and raises honey yield by 22 percent. Future work should test the protocol in hotter climates and evaluate alternative natural insulators.

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